EXCEPTION FLOW DOCUMENT

Restaurant Table Booking & Pre-Ordering App | Problem Statement #36

Purpose: Describe how the system responds when an exceptional condition interrupts the normal table-booking or pre-order workflow.

Scope note: The assigned problem statement explicitly identifies table availability, overbooking prevention, pre-order ticket queuing, preparation-time calculation and synchronized table status. The specific exception responses below are proposed handling rules for those scenarios; the problem statement does not prescribe exact error messages or recovery policies.

EF-01 – Selected table becomes unavailable

Trigger / Condition: The table is reserved by another diner before the current booking is confirmed, or the floor-plan status is stale.
System Response: The system informs the diner that the selected table is unavailable and refreshes the interactive floor plan.
Recovery / Resolution: The diner selects another open table. The system validates availability again and continues the booking flow. If no suitable table is selected, the diner may cancel the booking.

EF-02 – Overlapping reservation detected

Trigger / Condition: The system detects an existing reservation for the same table during an overlapping dining window.
System Response: The system rejects the conflicting reservation and does not mark the table as newly reserved.
Recovery / Resolution: The diner is asked to select another available table or a different time. The system must prevent the overlapping booking.

EF-03 – Pre-order menu item unavailable

Trigger / Condition: A selected menu item is no longer available when the diner reviews or confirms the pre-order.
System Response: The system identifies the unavailable item and prevents that item from being included in the kitchen pre-order ticket.
Recovery / Resolution: The diner removes the item or selects an available alternative, then reviews the updated pre-order before confirming.

EXCEPTION FLOWS – CONTINUED

EF-04 – Kitchen preparation timeline cannot be calculated

Trigger / Condition: The system cannot calculate the preparation timeline for the selected pre-order, for example because required preparation-time data is unavailable.
System Response: The system informs the diner that the preparation timeline cannot currently be calculated and does not display an estimated ready time as confirmed.
Recovery / Resolution: The diner may revise the pre-order or continue with the reservation without a pre-order, if the application permits. The system should only queue a pre-order ticket when the order details and timeline are available.

EF-05 – Reservation confirmation interrupted

Trigger / Condition: A connection or system interruption occurs while the diner is submitting the booking confirmation.
System Response: The system displays that confirmation could not be verified and checks the reservation record before accepting another submission.
Recovery / Resolution: If the reservation was saved, the system displays the existing confirmation. Otherwise, the diner can retry after availability is refreshed. This avoids duplicate reservations.

General handling rule: The system should preserve confirmed reservations, prevent overlapping table bookings, avoid duplicate confirmations, and queue kitchen tickets only for valid pre-orders.